Project Initialization and Planning Phase

Date	23-June-2026
Team ID	SWTID-2026-1377
Project Title	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to improve the evaluation of students' conceptual understanding using **Artificial Intelligence (AI)** and **Natural Language Processing (NLP)** techniques. Unlike traditional assessments that rely on keyword matching or objective questions, the system analyzes students' spoken explanations, identifies misconceptions and missing concepts, calculates a mastery score, and provides personalized feedback. The solution enhances learning effectiveness by enabling students to evaluate and improve their conceptual knowledge through voice-based interactions.

Project Overview	
Objective	To develop an AI-powered Voice-Based Concept Understanding Analyser that evaluates students' conceptual understanding through spoken explanations, identifies knowledge gaps, and provides personalized feedback with a mastery score.
Scope	The project captures voice or text input from students, converts speech into text, performs Natural Language Processing (NLP) and semantic similarity analysis, compares responses with a predefined knowledge base, detects misconceptions and missing concepts, calculates mastery scores, and generates personalized learning feedback through a Flask-based web application.
Problem Statement	
Description	Traditional assessment methods evaluate students using keyword matching or multiple-choice questions, which fail to accurately measure conceptual understanding. Students often memorize answers instead of developing deep conceptual knowledge, making it difficult to identify misconceptions and learning gaps.
Impact	Solving this problem improves conceptual learning, enhances examination and interview preparation, identifies knowledge gaps, promotes self-assessment, and provides personalized feedback that helps students strengthen their understanding of technical concepts.
Proposed Solution	

Approach	Develop a Flask-based web application that accepts voice or text input from students. Voice responses are converted into text using the Web Speech API, preprocessed using NLP techniques, compared with reference concepts using TF-IDF vectorization and cosine similarity, and evaluated to generate mastery scores, concept coverage, misconception detection, and personalized feedback.
Key Features	• Voice-based concept explanation using speech recognition. • Automatic speech-to-text conversion. • NLP-based semantic similarity analysis. • Detection of missing concepts and misconceptions. • Mastery score calculation and concept coverage analysis. • Personalized feedback and improvement suggestions. • Web-based interface for easy access and interaction.

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU/GPU specifications,	Standard CPU (No dedicated GPU required)
Memory	RAM specifications	Minimum **8 GB RAM**
Storage	Disk space for data, models, and logs	Approximately **2–5 GB** for source code, knowledge base, dependencies, and project files
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	Scikit-learn, NLTK, NumPy, Flask, JSON, Web Speech API
Development Environment	IDE	Visual Studio Code, Python 3.x, Git, GitHub
Data		
Data	Source, size, format	JSON-based concept knowledge base containing reference concepts, expected answers, keywords, and concept descriptions used for NLP-based evaluation.